
Computer Vision Lecture-1

Summary

Natural images are not just random 2D arrays of numbers. Although a computer stores an image as a grid of pixels, real-world images have structure because:

- * they are formed by light coming from a scene
- * reflection from surfaces
- * camera lens projection
- * the geometry of 3D objects.

Therefore, images follow the physics of light and the mathematics of projection (such as perspective geometry).

Natural images have properties resulting from image formation physics and math:

- * Objects create boundaries that cause sudden changes in color or brightness.
- * Surfaces are usually smooth, so intensity changes gradually.
- * Distant objects appear smaller due to perspective projection.
- * Lighting affects brightness but does not necessarily change shape.

Computer vision algorithms rely on these properties.

Solving computer vision problems requires using assumptions (priors). We cannot treat images as random data. Common priors include:

- * nearby pixels are similar
- * edges correspond to object boundaries
- * objects maintain structure under lighting changes.

Without these assumptions, vision tasks become impossible.

Scale analysis (zooming in or out) is necessary because objects appear at different sizes depending on distance. To detect both near and far objects, algorithms must search at multiple scales.

Matching patches (small pixel regions) between two images helps identify corresponding parts of the same object. This enables depth estimation, 3D location computation, and scene reconstruction. This principle is used in stereo vision.

Related problems include image restoration and image compression.

Image restoration improves degraded images through denoising and deblurring using priors like smoothness, randomness of noise, and edge preservation.

Image compression (such as JPEG, JPEG 2000, and MPEG) reduces file size by removing redundancy and exploiting image structure, such as similarity between nearby pixels and lower importance of high-frequency details.

Image filtering :

1. image restoration (denoising / deblurring)
2. image compression

Images contain noise because cameras and sensors are physical devices and light itself is random. Noise refers to unwanted random variations in pixel values.

Noise is not always harmful; in some cases, it can be beneficial. Controlled noise can improve visual perception making images appear smoother. It can enhance the detection of weak signals in certain systems. In machine learning, adding noise improves model robustness and prevents overfitting. Noise can also be used in digital forensics to identify the camera that captured an image. Additionally, controlled noise is used in computer graphics to generate realistic textures. **Thus, while uncontrolled noise degrades image quality, controlled noise can serve important practical purposes.**

One main source is sensor noise. Digital cameras measure light using sensors that count photons. Since photons arrive randomly, even a constant scene produces slightly fluctuating intensity values. This is called photon noise (shot noise). Hardware imperfections also cause dead pixels, which always output 0 (black) or 255 (white), appearing as bright or dark dots. In old film photographs, dust, scratches, and chemical aging introduce additional noise and grain.

An image is mathematically represented as a matrix of intensity values.

In grayscale images, each pixel has one brightness value.

In color images, each pixel has three values (Red, Green, Blue).

Typically, each value is stored in 1 byte, giving a range from 0 (black) to 255 (white).

To analyze noise mathematically, we assume:

- Noise is independent (one pixel's noise does not affect others).
- Noise follows a Gaussian distribution (bell-shaped curve: small variations are common, large ones are rare).

Gaussian noise appears grainy (salt and pepper noise) and is common because many small random effects combine, which statistically leads to a Gaussian distribution.

Noise reduction relies on a key property of natural images: nearby pixels usually belong to the same object and have similar intensities, while noise is random.

Mean filtering reduces noise by replacing each pixel with the average of its neighbors (e.g., in a 3×3 window, sum 9 values and divide by 9). This smooths the image and reduces noise because random variations cancel out. However, it also blurs edges, since sharp changes are averaged.

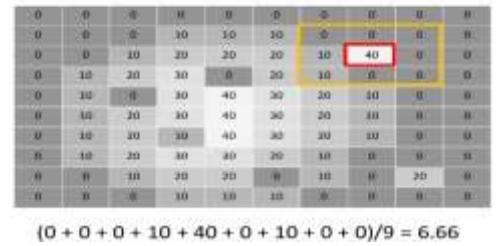
Human eyes can not detect less than 40 levels of each color

If we used a wider grid than 3×3 it will make the image smoother (it will lose some details)

Slide 10 :

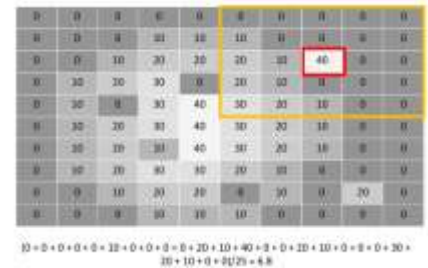
- * **Concept:** Introduces the 3x3 **kernel** (the orange box).
- * **Process:** To find the new value for a pixel, you take the average of it and its 8 immediate neighbors.
- * **Result:** A sharp "hot spot" (value 40) is diluted down to a much lower value (6.66).

Mean filtering



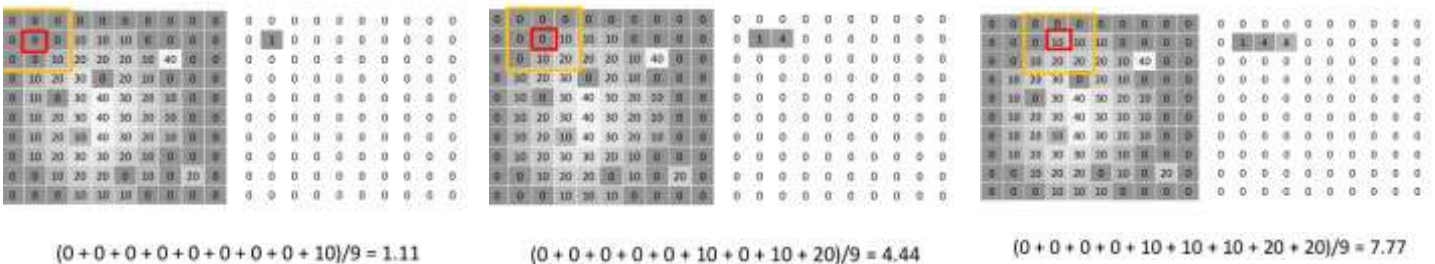
Slide 11: Changing the Scale (5x5)

- * **Concept:** Shows how a larger **kernel** (5 x 5) works.
- * **Process:** You now average 25 pixels instead of 9.
- * **Impact:** Larger filters create a much stronger blur because they pull information from a wider area.



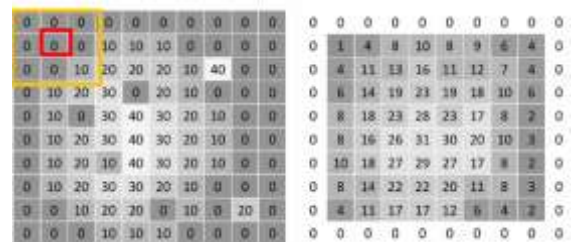
Slide 12 , 13 , 14: Starting the "Convolution"

- * **Concept:** Shows the start of the actual image processing and Demonstrates the "sliding" nature of the filter.
- * **Process:** The filter starts at the first possible position (top-left) then shifts one pixel to the right at a time, recalculating the average for each new position.
- * **Visual:** The right-hand grid represents the new image, with the calculated value (1, 4 then 8) appearing.



Slide 15: The Finished Product

- * **Concept:** The final result of the filtering operation.
- * **Comparison:** The original image has sharp edges and noise; the new image is smoothed.
- * **Key Limitation:** Notice the "Border Effect" , because the 3x3 box couldn't fit over the very edges of the image, the outer boundary of the output grid remains zero (black).



Mean Filtering (3×3 Kernel)

$$S[f](m,n) = \frac{1}{9} \sum_{i=-1}^1 \sum_{j=-1}^1 f(m+i, n+j)$$

Where:

- $S[f](m,n)$ is the output (filtered) pixel value at row m , column n
 - (m, n) is the coordinate of the specific pixel currently being calculated
 - $f(m+i, n+j)$ = original pixel values in the 3×3 neighborhood
 - $\sum_{i=-1}^1 \sum_{j=-1}^1$ = sum over all neighbors (including the center pixel)
 - Division by 9 = averaging step to compute the mean
 - i makes us move vertically and j makes us moves horizontally
 - if i and j are equal to 1 then we are at the right bottom of the center , if i is equal to -1 and j is equal to 1 then we are at the top right and so on....
-

General Weighted Kernel (2k+1 × 2k+1)

$$S[f](m,n) = \sum_{i=-k}^k \sum_{j=-k}^k w(i,j) f(m+i, n+j)$$

Where:

- $S[f](m,n)$ = output (filtered) pixel
 - $f(m+i, n+j)$ = input pixel in the local neighborhood
 - $w(i,j)$ = weight of each neighbor in the kernel
 - k = kernel “radius” (e.g., $k=1 \rightarrow 3 \times 3$ kernel, $k=2 \rightarrow 5 \times 5$ kernel)
-

Weighted mean filtering generalizes this idea by assigning different weights to neighboring pixels using a kernel. The general formula sums weighted neighboring pixels.

Kernel size = $2k + 1$ (e.g., $k = 1 \rightarrow 3 \times 3$, $k = 2 \rightarrow 5 \times 5$).

* Different weight patterns produce different effects:

- Equal weights → smoothing (mean filter)
- Larger center weight → sharpening
- Positive/negative patterns → edge detection

Filter	Best for	Edge Preservation	Notes / Use Case
Mean	Low Gaussian noise, slight skew	Poor	Simple smoothing
Median	Salt-and-pepper, high variance, skewed	Good	Robust to outliers
Mode	Quantized images, repeated values	Moderate	Binary or posterized images
Weighted Mean	Moderate Gaussian noise, small texture	Moderate	Center-weighted smoothing

Cross-correlation vs Convolution

- **Cross-correlation (general weighted kernel):**

$$S[f](m, n) = \sum_{i=-k}^k \sum_{j=-k}^k w(i, j) \otimes f(m + i, n + j)$$

- **Convolution:** widely used to transfer information from one space to another

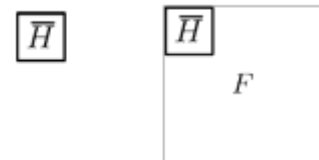
$$S[f](m, n) = \sum_{i=-k}^k \sum_{j=-k}^k w(i, j) * f(m - i, n - j)$$

- Difference: convolution flips the kernel horizontally and vertically before applying, cross-correlation does not.

Convolution

Slide 23 Core Concept

- * **F (The Input):** represents the original image or data grid.
- * **H(The Kernel/Filter):** This is a small matrix of numbers designed to find specific features (like edges or blur). The bar over the H indicates it has been **flipped** 180 degrees, which is the formal mathematical requirement for convolution.



The Action

The diagram shows the being placed over a specific patch of the image. To get a single output value:

1. **Multiply** each number in the filter by the pixel it is sitting on.
2. **Add** all those results together.
3. **Slide** the filter to the next position and repeat.

Scaling

- * If the input image is scaled by a constant a :

$$f'(m,n) = a f(m,n) \Rightarrow (w \otimes f')(m,n) = a (w \otimes f)(m,n)$$

- * **Simplified:** $f' = af \Rightarrow w \otimes f' = a (w \otimes f)$

Example: Doubling the brightness of an image doubles the filtered output.

Superposition (Additivity)

- * If the input is a linear combination of two images f and g :

$$f' = af + bg \Rightarrow w \otimes f' = a (w \otimes f) + b (w \otimes g)$$

- * If the kernel is a linear combination of two kernels w and v :

$$w' = aw + bv \Rightarrow w' \otimes f = a (w \otimes f) + b (v \otimes f)$$

This allows breaking down complex signals into simpler components.

2. Shift Invariance (Translation Invariance)

- Definition of a shift in the input: $f'(m,n) = f(m - m_0, n - n_0)$

- Mathematical proof that convolution is shift-invariant:

$$\begin{aligned} (w \otimes f')(m,n) &= \sum_{i=-k}^k \sum_{j=-k}^k w(i,j) f'(m+i, n+j) \\ &= \sum_{i=-k}^k \sum_{j=-k}^k w(i,j) f(m+i-m_0, n+j-n_0) \\ &= (w \otimes f)(m-m_0, n-n_0) \end{aligned}$$

- **Key Insight:** Shift then convolve = Convolve then shift
This means the output of convolution does not depend on the absolute location of a feature in the image.
- Correlation (and convolution) ensures that **filters detect features regardless of their position**, which is exactly what shift invariance means.

Basic Filter Examples in Image Processing

1. Mean Filter (Blurring)

- **Kernel:** $k = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

- **Effect:** Calculates the average value of a pixel and its 8 neighbors.
- **Result:** Smooths transitions between pixels → blurred image.

Mathematical formula: $g(m,n) = \frac{1}{9} \sum_{i=-1}^1 \sum_{j=-1}^1 f(m+i, n+j)$

2. Identity Filter

- * **Kernel:** $k = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

- * **Effect:** Keeps the center pixel and ignores neighbors.
- * **Result:** Output is identical to the original image.

Mathematical formula: $g(m,n) = 1 \cdot f(m,n) + 0 \cdot \text{neighbors} = f(m,n)$

3. image reconstructing : Sharpening

Sharpening **accentuates edges and fine details** by adding back what blurring removes.

Step 1: Extract Details $\text{Edges} = f - f_{blur}$

Where f_{blur} is the blurred version of the image (e.g., via mean filter).

Step 2: Sharpening Equation

$$f_{sharp} = f + \alpha(f - f_{blur})$$

- α = scaling factor controlling the strength of sharpening

Rewriting as a single linear operation:

$$f_{sharp} = (1 + \alpha)f - \alpha f_{blur}$$

Step 3: Combined Sharpening Kernel

- Using linearity, the **Identity kernel** w and **Mean kernel** v can be combined:

$$w_{sharp} = (1 + \alpha)w - \alpha v$$

- Example: If $\alpha = 1$, the **center of the identity kernel** becomes 2, and the mean kernel is subtracted.
- **Effect:** Highlights edges while keeping the original image structure

MCQ

1. Why are natural images not just random 2D arrays of numbers?
 - A. Because images are always black and white
 - B. Because images are influenced by the physics of light and 3D geometry
 - C. Because computers cannot store random data
 - D. Because all pixels have the same value

Answer: B
2. Which property of natural images helps detect object boundaries?
 - A. Objects create boundaries causing sudden changes in color or brightness
 - B. Camera sensors eliminate noise
 - C. Distant objects appear larger
 - D. Lighting is uniform across the scene

Answer: A
3. Why is scale analysis important in computer vision?
 - A. To remove sensor noise
 - B. To detect objects of different sizes due to distance
 - C. To increase pixel intensity
 - D. To convert images to grayscale

Answer: B
4. Matching small patches between two images helps to:
 - A. Reduce JPEG compression artifacts
 - B. Identify corresponding parts of the same object
 - C. Convert color images to grayscale
 - D. Increase image resolution

Answer: B
5. What is the main goal of image restoration?
 - A. Detect edges
 - B. Improve degraded images through denoising and deblurring
 - C. Convert 2D images to 3D
 - D. Reduce file size

Answer: B
6. Image compression works by:
 - A. Removing redundancy and exploiting image structure
 - B. Eliminating object boundaries
 - C. Adding noise to enhance textures
 - D. Increasing brightness

Answer: A

7. Which of the following is a source of noise in digital images?

- A. Perspective projection
- B. Sensor imperfections and photon noise
- C. Camera lens zoom
- D. Object boundaries

Answer: B

8. Photon noise (shot noise) occurs because:

- A. Pixels always output 0 or 255
- B. Photons arrive randomly even in constant scenes
- C. Objects are far away
- D. Sensors smooth nearby pixel values

Answer: B

9. What is the main characteristic of Gaussian noise?

- A. Large variations are common, small variations are rare
- B. Small variations are common, large ones are rare
- C. Noise appears only in color images
- D. Noise can only occur due to lens defects

Answer: B

10. Why is mean filtering used in image processing?

- A. To detect edges with high accuracy
- B. To replace each pixel with the average of its neighbors, reducing noise
- C. To compress the image for storage
- D. To increase the brightness of pixels

Answer: B

11. What is a drawback of using mean filtering?

- A. It blurs edges due to averaging sharp changes
- B. It cannot reduce noise
- C. It increases photon noise
- D. It removes all object boundaries

Answer: A

12. How are grayscale and color images represented mathematically?

- A. Grayscale: 3 values per pixel; Color: 1 value per pixel
- B. Grayscale: 1 value per pixel; Color: 3 values per pixel (R, G, B)
- C. Grayscale: 1 value per pixel; Color: 4 values per pixel (R, G, B, E)
- D. Both have only 1 value per pixel

Answer: B

13. What is the typical range of pixel intensity values in 1-byte storage?

- A. 0–512
- B. 1–255
- C. 0–255
- D. 0–1023

Answer: C

14. Controlled noise in images can be beneficial for:

- A. Reducing pixel intensity
- B. Destroying image details
- C. Improving visual perception
- D. Increasing sensor imperfections

Answer: C

15. Why do nearby pixels usually have similar intensities?

- A. Because distant objects appear smaller
- B. Because they usually belong to the same object
- C. Because noise is always uniform
- D. Because cameras store the same value for neighbors

Answer: B

16. Increasing the size of the filtering grid (e.g., wider than 3×3) will:

- A. Make the image smoother but lose some details
- B. Remove all noise without affecting edges
- C. Make the image sharper
- D. Increase the number of colors in the image

Answer: A

17. What does human eye limitation suggest about color detection?

- A. Humans cannot detect less than 40 levels of each color
- B. Humans can detect unlimited levels of color
- C. Humans cannot detect grayscale variations
- D. Humans can detect only 2 levels of each color

Answer: A

18. In a 3×3 mean filter, how is the new value of a pixel calculated?

- A. By averaging the pixel with its 8 immediate neighbors
- B. By multiplying all neighboring pixels
- C. By taking the maximum value of its 8 neighbors
- D. By subtracting the pixel value from the neighbors

Answer: A

19. What happens to a sharp "hot spot" value when using a 3×3 mean filter?

- A. It is diluted to a lower value
- B. It increases sharply
- C. It remains the same
- D. It becomes negative

Answer: A

20. How does a larger kernel, such as 5×5 , affect the image compared to a 3×3 kernel?

- A. It creates a stronger blur
- B. It removes all color information
- C. It sharpens edges
- D. It reduces image size

Answer: A

21. During convolution, what is meant by the “sliding” nature of the filter?

- A. The filter shifts across the image
- B. The filter moves randomly
- C. The filter only affects the edges
- D. The filter rotates around the image

Answer: A

22. What is the main limitation of mean filtering regarding image borders?

- A. Outer boundaries may remain zero due to the filter not fitting fully
- B. Borders become sharper than the center
- C. Borders are enhanced automatically
- D. Borders are ignored completely in analysis

Answer: A

23. How does a general weighted kernel differ from a simple mean filter?

- A. It assigns different weights to each neighboring pixel
- B. It ignores the center pixel
- C. It always increases the center pixel value
- D. It only uses the four diagonal neighbors

Answer: A

24. What is the kernel size formula for a weighted filter?

- A. k^2
- B. $k + 2$
- C. $2k + 1$
- D. $2k$

Answer: C

25. Why is weighted mean filtering preferred over simple mean in some cases?

- A. It reduces the number of pixels
- B. It preserves edges
- C. It increases blur uniformly
- D. It ignores noise completely

Answer: B

26. Which filter is robust to outliers in the image?

- A. Weighted mean
- B. Mode
- C. Median
- D. Mean

Answer: C

27. What effect does assigning a larger weight to the center pixel in a kernel produce?

- A. Edge removal
- B. Sharpening
- C. Noise addition
- D. Blurring

Answer: B

28. For which type of images is the mode filter most suitable?

- A. Images with Gaussian noise
- B. High-resolution photographs
- C. Quantized or posterized images with repeated values
- D. Videos

Answer: C

29. Which filter is considered simple but provides poor edge preservation?

- A. Weighted mean
- B. Median
- C. Mode
- D. Mean

Answer: D

30. What does the "k" in a $2k+1 \times 2k+1$ kernel represent?

- A. Kernel radius (distance from center to edge)
- B. Image width
- C. Number of neighbors to ignore
- D. Number of color channels

Answer: A

31. What is the key difference between cross-correlation and convolution?

- A. Convolution flips the kernel horizontally and vertically, cross-correlation does not
- B. Convolution only works on color images
- C. Cross-correlation flips the kernel horizontally and vertically, convolution does not
- D. Both always flip the kernel

Answer: A

32. How is a single output value computed during convolution?

- A. Multiply each kernel element by the corresponding pixel and sum the results
- B. Add all input pixel values without multiplying by the kernel
- C. Subtract the kernel from the image patch
- D. Take the maximum of all multiplied values

Answer: A

33. If the input image is scaled by a constant factor a , how does the convolution output change?

- A. The filtered output is also multiplied by a
- B. The output is squared
- C. It becomes independent of the scaling
- D. The kernel must be recomputed

Answer: A

34. If a kernel is a linear combination of two kernels, what is the effect on convolution?

- A. The output is always zero
- B. Convolution is the linear combination of the convolutions with each kernel
- C. Convolution is the same as applying one kernel only
- D. Convolution cannot be applied

Answer: B

35. What does shift invariance (translation invariance) mean in convolution?
- A. Shifting the input image before or after convolution gives the same result
 - B. Output depends on the absolute location of features
 - C. Features cannot be detected after shifting
 - D. Only applicable for 3×3 kernels

Answer: A

36. Why is shift invariance important in image processing?
- A. It allows filters to detect features regardless of their position
 - B. It only works for grayscale images
 - C. It removes noise automatically
 - D. It increases the image resolution

Answer: A

37. For a 3×3 mean filter kernel, what is the weight of each pixel?
- A. 1/5
 - B. 1/3
 - C. 1/8
 - D. 1/9

Answer: D

38. What is the main effect of applying a mean filter to an image?
- A. Doubling pixel intensity
 - B. Smoothing transitions
 - C. Detecting edges only
 - D. Sharpening edges

Answer: B

39. In convolution notation, $S_f = \sum \sum w(i,j) * f(m-i, n-j)$. Why are the indices negative in the formula?
- A. To flip the kernel horizontally and vertically
 - B. To rotate the image
 - C. To ignore border pixels
 - D. To normalize the output

Answer: A

40. If you double the brightness of an image (multiply by 2), what happens to the output of a linear convolution?
- A. It stays the same
 - B. The output doubles
 - C. The output halves
 - D. The output becomes zero

Answer: B

41. Which statement is true regarding cross-correlation?
- A. It does not flip the kernel before applying it
 - B. It always flips the kernel
 - C. It is always non-linear

D. It only works on color images

Answer: A

42. During convolution, what is the first step when placing the kernel on the image?

A. Multiply each kernel element by the pixel it overlaps

B. Flip the image

C. Divide each pixel by the kernel sum

D. Shift the kernel diagonally

Answer: A

43. What is the main purpose of the identity filter?

A. Sharpen edges

B. Detect corners

C. Keep the center pixel and ignore neighbors

D. Blur the image

Answer: C

44. What is the output of an identity filter applied to an image?

A. An edge-detected image

B. A blurred version of the image

C. An identical copy of the original image

D. An inverted image

Answer: C

45. What is the kernel for a 3×3 identity filter?

A. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

B. $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

Answer: A

46. □ What is the first step in image sharpening?

A. Add Gaussian noise

B. Multiply by a scaling factor

C. Extract details by subtracting the blurred image from the original

D. Apply an identity filter

Answer: C

47. □ Which formula represents the sharpening operation?

A. $f_{\text{sharp}} = f \times f_{\text{blur}}$

B. $f_{\text{sharp}} = f / f_{\text{blur}}$

C. $f_{\text{sharp}} = f + \alpha(f - f_{\text{blur}})$

D. $f_{\text{sharp}} = f_{\text{blur}} - f$

Answer: C

48. □ In the sharpening equation $f_{\text{sharp}} = f + \alpha(f - f_{\text{blur}})$, what does α control?

A. The blur strength

B. The strength of the sharpening effect

C. The color saturation

D. The brightness of the image

Answer: B

49. □ How can sharpening be rewritten as a single linear operation?

A. $f_{\text{sharp}} = \alpha f + f_{\text{blur}}$

- B. $f_{\text{sharp}} = f / \alpha$
- C. $f_{\text{sharp}} = (1+\alpha)f - \alpha f_{\text{blur}}$
- D. $f_{\text{sharp}} = f - f_{\text{blur}}$

Answer: C

50. ☐ How can the identity kernel and mean kernel be combined for sharpening?

- A. $w_{\text{sharp}} = w \times v$
- B. $w_{\text{sharp}} = (1+\alpha)w - \alpha v$
- C. $w_{\text{sharp}} = w / v$
- D. $w_{\text{sharp}} = w + v$

Answer: B

51. ☐ If $\alpha = 1$ in the combined sharpening kernel, what happens to the center value of the identity kernel?

- A. It becomes 0
- B. It becomes -1
- C. It becomes 2
- D. It becomes 1

Answer: C

52. ☐ What is the effect of applying a sharpening filter with $\alpha = 1$?

- A. Blurs the image
- B. Highlights edges while keeping original image structure
- C. Darkens the image uniformly
- D. Removes noise completely

Answer: B

53. ☐ Why is the identity filter important in sharpening?

- A. It increases the kernel size
- B. It provides the reference original image to preserve structure
- C. It reduces noise
- D. It adds Gaussian noise

Answer: B

54. ☐ Sharpening can be considered a combination of which two operations?

- A. Identity filtering and blurring subtraction
- B. Cross-correlation and scaling
- C. Median filtering and edge detection
- D. Mean filtering and color inversion

Answer: A

55. ☐ Which of the following best explains why convolution flips the kernel while cross-correlation does not?

- A. Flipping aligns the kernel mathematically to transfer information correctly between spaces
- B. Flipping ensures that shift invariance is violated
- C. Flipping increases kernel size for better smoothing
- D. Flipping reduces image noise

Answer: A

56. ☐ Consider a 5×5 weighted kernel with non-uniform weights. Which property of natural images justifies giving the center pixel a higher weight than its neighbors?

- A. Nearby pixels tend to belong to the same object, so emphasizing the center preserves details
- B. Distant objects appear smaller
- C. Photon noise dominates in the corners
- D. Image scaling reduces edge intensity

Answer: A

57. ☐ Which mathematical principle allows complex filtering operations to be decomposed into simpler convolutions, such as separating a sharpening operation into identity and mean kernels?

- A. Gaussian noise distribution
- B. Superposition (linearity)
- C. Shift invariance
- D. Correlation coefficient

Answer: B

58. ☐ When applying a 3×3 mean filter to an image with a single bright “hot spot,” why does the spot’s intensity reduce significantly?

- A. Because averaging incorporates surrounding lower-intensity pixels
- B. Because edges are enhanced automatically
- C. Because the kernel ignores the center pixel
- D. Because the filter flips the image

Answer: A

59. ☐ Which of the following is a key limitation of mean filtering when applied repeatedly or with large kernels?

- A. It enhances edges excessively
- B. It causes a strong blur and may lose important high-frequency details
- C. It makes the image shift-invariant
- D. It cannot be applied to grayscale images

Answer: B

60. ☐ In sharpening an image using $f_{\text{sharp}} = f + \alpha(f - f_{\text{blur}})$, what is the role of the term $(f - f_{\text{blur}})$?

- A. It isolates the image details or edges that were lost during blurring
- B. It scales the brightness globally
- C. It represents the smooth regions of the image
- D. It increases image noise

Answer: A

61. ☐ Which filter combination would best highlight edges while preserving the original image structure?

- A. Identity filter + weighted subtraction of blurred image
- B. Cross-correlation with a uniform kernel
- C. Mean filter only
- D. Median filter only

Answer: A

62. ☐ Considering the principles of convolution, scaling, and superposition, which statement correctly describes the output when both the input and kernel are linear combinations of multiple images and filters?

- A. The output equals the linear combination of the convolutions of each individual image with each individual kernel
- B. Only the first image in the combination contributes to the output
- C. Linear combination breaks shift invariance
- D. The output cannot be predicted without full recomputation

Answer: A